

|   |     |                                                                                                                 |    |     |
|---|-----|-----------------------------------------------------------------------------------------------------------------|----|-----|
|   |     | $L = 1.1 \text{ m}$ .....                                                                                       | A1 | [5] |
| 5 | (a) | (i) arrow from B towards A.....                                                                                 | B1 |     |
|   |     | (ii) $E = V/d$<br>$= 450/(9.0 \times 10^{-2})$ .....                                                            | C1 |     |
|   |     | $= 5.0 \times 10^3 \text{ N C}^{-1}$ (accept 1 sig. fig) .....                                                  | A1 | [3] |
|   | (b) | (i) energy = $qV$ or $Eqd$ .....                                                                                | C1 |     |
|   |     | $= 1.6 \times 10^{-19} \times 450$ .....                                                                        | A1 |     |
|   |     | $= 7.2 \times 10^{-17} \text{ J}$ .....                                                                         | A0 |     |
|   |     | (ii) $E_k = \frac{1}{2}mv^2$<br>$7.2 \times 10^{-17} = \frac{1}{2} \times 9.1 \times 10^{-31} \times v^2$ ..... | C1 |     |
|   |     | $v = 1.26 \times 10^7 \text{ m s}^{-1}$ .....                                                                   | A1 | [4] |
|   | (c) | line from origin, curved in correct direction but not 'level out' .....                                         | B1 | [1] |
| 6 | (a) | (i) 26 protons .....                                                                                            | B1 |     |